The effect of the 1994 franc CFA devaluation on the WAEMU and CEMAC zones

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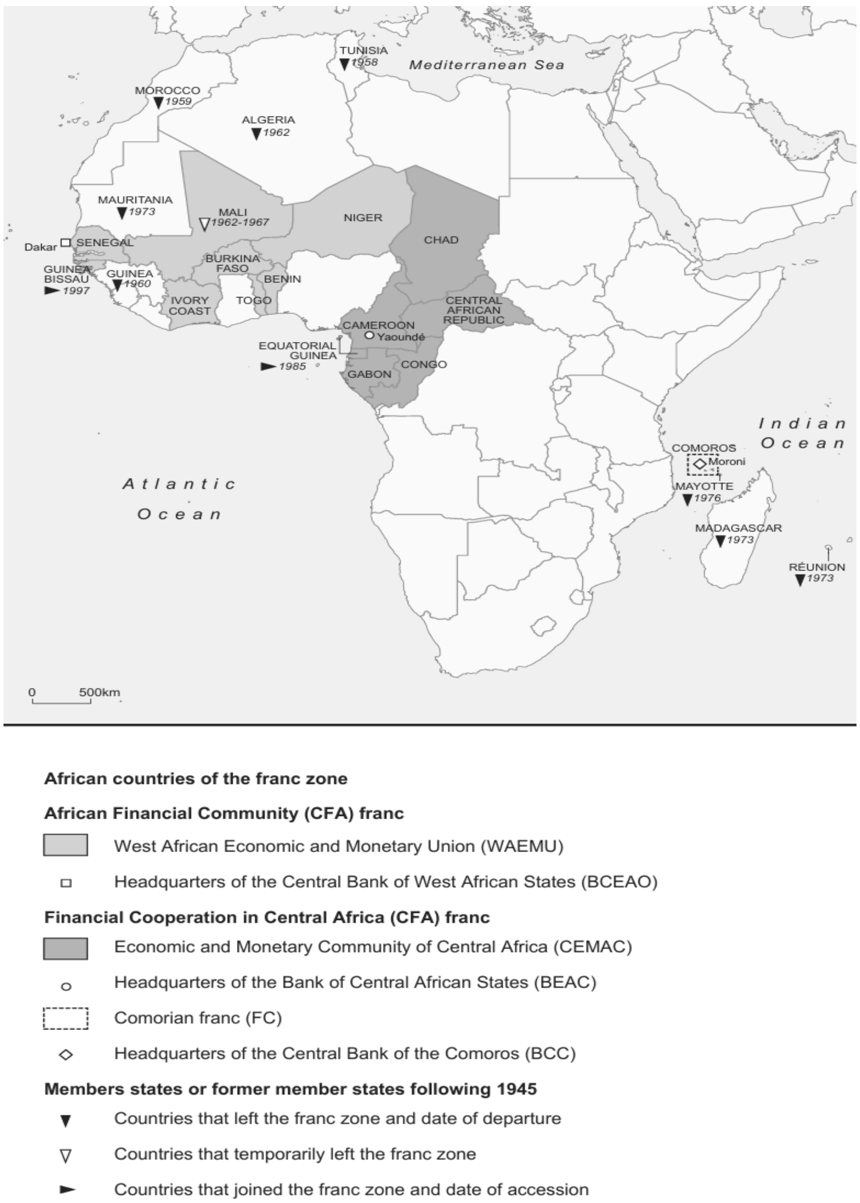

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[1] Pigeaud, Fanny, Ndongo Samba Sylla, and Thomas Fazi. Africa's Last Colonial Currency: The CFA Franc Story. 1 online resource (xiii) : map vols. London: Pluto Press, 2020.

Table of Contents

Central Question: How did the 1994 French devaluation of the CFA franc affect the economies of the monetary zones operating under the BEAC and BCEAO?

Abstract

On January 12th, 1994, France devalued the CFA franc relative to the French franc, raising the parity rate between the two currencies from 50 CFA francs per French franc to 100 francs per French franc. This research project examines the devaluation's economic impacts on monetary zones operating under the The Banque Centrale des États de l'Afrique de l'Ouest (BEAC) and the Banque des États de l'Afrique Centrale (BCEAO), two central banks for regional countries in Africa. In order to properly examine the effects of the devaluation's our study collected data on 4 different economic variables. Our data was then interpreted through performing Welch T-Tests comparing pre- and post- means across all CFA countries. The direct results were then analyzed through the assumptions of the Mundell-Fleming model. Our project concluded that the economic effect of the devaluation on CFA countries challenged the base assumptions of the Mundell-Fleming model and highlighted the importance of governmental structure when discussing devaluation impacts.

I. Introduction

"France is there as a friend", said the French economic minister Michel Sapin about the African Franc Zone in 2017, the world's oldest monetary union.[2] Why was such a declaration made even though the last French colony in Africa became independent 40 years earlier? It appears that independence was never fully achieved for the 14 members of the Franc Zone, named after France's currency prior to its transition to the euro in 1999. This group of countries divided into a West and Central African Central Bank still use the CFA franc. The currency was introduced on December 25th, 1945 after the Bretton Woods Conference that reshaped the international financial order.[3] Largely managed from Paris, its banknotes are printed in the French metropole and hand over the monetary autonomy of its member states

[2] 'Le Franc, malgré son nom, est la monnaie des Africains'. Jeune Afrique. April 14, 2017.
[3] France. "Journal Officiel de la République Française." Lois et Décrets, December 26, 1945.

to France. To be clear, these banknotes cannot be exchanged across countries of the Franc Zone. If an individual from the Ivory Coast traveling to Cameroon wants to use his CFA francs, he must first convert Ivorian banknotes to euros and then convert euros to Cameroonian banknotes. For a long time, debates surrounding the relevance and purpose of the Franc Zone were intentionally hidden from the public by the French authorities' disinterested attitude, when in reality it is key to private and government interests to the detriment of local African populations.

It is important to provide economic context pertaining to the structure of this monetary system. Members of the zone are subordinated to the trade and financial arrangements of France,[4] mainly because the exchange rate of the CFA franc is fixed to the euro (the French franc prior to 1999).[5] Typically, a floating exchange rate allows a country to decrease or increase the value of its domestic currency relative to others to better reflect economic conditions. However, since the parity between the Euro and the CFA franc is set by the French government, the exchange rate can be manipulated to reflect French interests. For example, the CFA franc was overvalued relative to the French franc at the time of its creation. This made goods from the metropole cheaper than domestic ones, encouraging colonies to import from France. The overvaluation also made the Franc Zone's domestic goods unattractive on global markets, meaning they had to turn to the metropole to export materials. This decision to overvalue the CFA franc was made to repair the French economy, which had suffered during World War II, at the expense of the economies in the Franc Zone. It is estimated that the livelihoods of 500,000 people in the Metropole depends on the Franc Zone's economy.[6] The free movement of capital allows for an extractive system where French companies can easily redirect profits to the metropole without needing to invest in the country of production.

[4] Speech by Kwame Nkrumah at the summit of the Organisation of African Unity.
[5] The exchange rate is the value at which a currency can be traded for another. Today, 1 euro equals 656 CFA francs. Although the economic details of the Franc Zone are intricate, all of its conditions have something in common: they make members dependent on France and the Euro for their economic conditions.
[6] 'La zone franc: survivance du passé ou promesse d'avenir'. La Croix, February 17, 1960

French officials tend to praise the currency as a gateway to economic stability, while African officials often complain about the lack of monetary autonomy and sovereignty the Franc Zone perpetuates. These contrasts can best be understood in the context of overwhelming evidence that presents cooperation agreements between France and former colonies as inherently favoring the metropole. French authorities publicly praise the ‘independence’ and ‘stability’ of the Franc Zone while privately acknowledging the economic benefit the metropole reaps from the arrangement. This discourse serves to disguise the system’s colonial origins. The metropole’s grip over the West and Central African central banks through its representatives that hold an implicit right to veto any important monetary decision render the zone's member states’ claims to sovereignty shaky.

Given the problematic colonial context of the monetary zone, we thought it valuable to analyze the effects of the 1994 devaluation of the Franc Zone’s economy relative to the CFA franc. We tested whether the predictions of the Mundell Flemming model for a small country with a small open economy under a fixed exchange rate regime apply to this peculiar context. The model predicts that:

(1) The devaluation of a currency leads to an increase in net exports since domestic goods become cheaper and foreign goods become more expensive (increase in exports, decrease in imports).

(2) The devaluation is expected to increase real per Capita GDP growth Rate by boosting net exports through improved price competitiveness.

(3) Currency devaluation reduces overall household final consumption due to the increased costs of imports.

(4) While the effect of currency devaluation on general government final consumption is unclear, it can provoke fiscal policy responses worth researching.

II. Literature Review

The Mundell-Fleming model suggests that currency devaluation in a small open economy increases net exports by making domestic goods more competitive abroad and by raising import prices. This, in turn, should promote GDP growth by increasing external demand, even as higher import costs may lower domestic consumption. This theoretical framework underpins both the IMF's and France's support for the 1994 CFA franc devaluation (from 50 to 100 CFA francs per French franc). The adjustment sought to restore competitiveness in the BCEAO (West African Economic and Monetary Union) and BEAC (Central African Economic and Monetary Community) economies, after years of struggling with an overvalued exchange rate.

In general, early post-devaluation assessments, such as those by Clément (1995) and Clément et al. (1996), supported the effectiveness of the policy.[7] They reported a recovery in growth led by export sectors, particularly agriculture, along with inflationary pressures that were less severe than expected (both in magnitude and duration). Similarly, Van den Boogaerde and Tsangarides (2005) confirmed improvements in growth, inflation control, and competitiveness within the first four years after the devaluation.[8]

However, more recent studies, such as Bouvet, Bower, and Jones (2022), have challenged these earlier findings. Using the Augmented Synthetic Control Method (ASCM), they find that the 1994 devaluation failed to generate a statistically significant improvement in GDP per capita across the CFA franc zone, with the exception of Mali. The divergence between the theoretical expectations, initial research, and more recent findings may be explained by factors such as political instability, including the

[7] Carmen M. Reinhart, "Devaluation, Relative Prices, and International Trade: Evidence from Developing Countries," *IMF Staff Papers* 42, no. 2 (1995): 290–312.
[8] Pierre Van den Boogaerde and Charalambos Tsangarides, *Ten Years After the CFA Franc Devaluation: Progress Toward Regional Integration in the WAEMU*, IMF Working Paper 05/145 (Washington, DC: International Monetary Fund, 2005).

1996 attempted coup d'état in the Central African Republic, as well as the weak implementation of IMF-mandated structural reforms.[9]

Additionally, Reinhart (1995) is referenced in Bouvet et al. (2022) as a framework for evaluating the success of a devaluation. The criteria include: (1) nominal devaluation must result in real devaluation, and (2) trade flows must respond to changes in relative prices.[10] Bouvet et al. observe that while the first condition was met, as indicated by an adjustment in the real exchange rate, the second was not. This shortcoming is attributed to the CFA zone's reliance on price-inelastic commodity exports and low export-to-GDP ratios, which limited the devaluation’s potential to stimulate growth.

Dillner (2022) further examines the devaluation’s effects on exports. He finds that exports as a percentage of GDP increased by an average of 5.03 percentage points following the devaluation. However, this rise was driven primarily by higher export prices in CFA francs rather than by growth in export volumes. Moreover, export volumes responded sluggishly, and exports measured in USD showed no significant increase.[11] This implies a high exchange-rate pass-through, where CFA franc prices rose to offset the devaluation while USD prices remained stable, reflecting the CFA zone's structural limitations as price-takers of undifferentiated commodities with inelastic production capacity.

Our study expands on previous research by looking at a broader set of macroeconomic indicators, including net exports, real GDP growth, household consumption, and government spending, across both CFA zones. While previous research focused on GDP per capita or trade volumes in isolation, our approach provides a broader overview of the devaluation's macroeconomic impact. This is relevant given that the CFA franc system has long been a topic of debate, with differing views on whether it promotes stability or limits economic zone autonomy. We hope to provide another perspective on the diverse

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[9] Florence Bouvet, Jonathan Bower, and Byron F. Jones, “Revisiting the CFA Franc Devaluation Using the Augmented Synthetic Control Method” (Working paper, 2022).
[10] Carmen M. Reinhart, “Devaluation, Relative Prices, and International Trade: Evidence from Developing Countries,” *IMF Staff Papers* 42, no. 2 (1995): 290–312.
[11] Matthew Dillner, “Export Outcomes After the 1994 CFA Franc Devaluation: A Synthetic Control Analysis” (Working paper, 2022).

economic responses to the 1994 devaluation at the regional level by connecting our findings to the Mundell-Fleming model's predictions and using both t-tests and visualizations.

III. Methodology

A. Variables

In our research, we seek to evaluate the effect of the 1994 devaluation on the following economic variables, drawn from this Dataset:[12]

(1) Net exports (current US$)

(2) Real GDP Growth (annual %)

(3) Household final consumption expenditure (% of GDP)

(4) Government final consumption expenditure (% of GDP)

B. Units of analysis

We observe how these variables are affected in the monetary zones operating under the two central banks that use the CFA franc:

(1) The Central Bank of West African States (BCEAO), consisting of countries in the West African Economic and Monetary Union (WAEMU) which includes Senegal, Mali, Ivory Coast, Burkina Faso, Benin, Togo, Guinea Bissau, and Niger.

(2) The Bank of Central African States (BEAC), consisting of countries in the Economic and Monetary Community of Central Africa (CEMAC) which includes Chad, Cameroon, the Central African Republic, Gabon, Equatorial Guinea, and the Congo.[13]

C. Statistical Method and Visualizations

See RMD file attached to see the coding structure for the following operations:

(1) We calculate average levels of each key indicator by zone and period.

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[12] "African Economy from 1980 to 2022." https://www.kaggle.com/datasets/mahmoudsaeed99/african-economy-from-1980-to-2022.

[13] It is important to note that two members of the Franc Zone, Equatorial Guinea and Guinea Bissau, are former Spanish and Portuguese colonies respectively. Consequently, the context in which they came to be brought into the zone differs.

(2) We plot a time series of Net Exports, Real GDP Growth, Household Final Consumption Expenditure, and General Government Final Consumption Expenditure, marking the devaluation year with a dashed red line.

(3) To formally assess the devaluation effect, we perform Welch t-tests comparing pre- and post-means for each variable across all CFA countries.

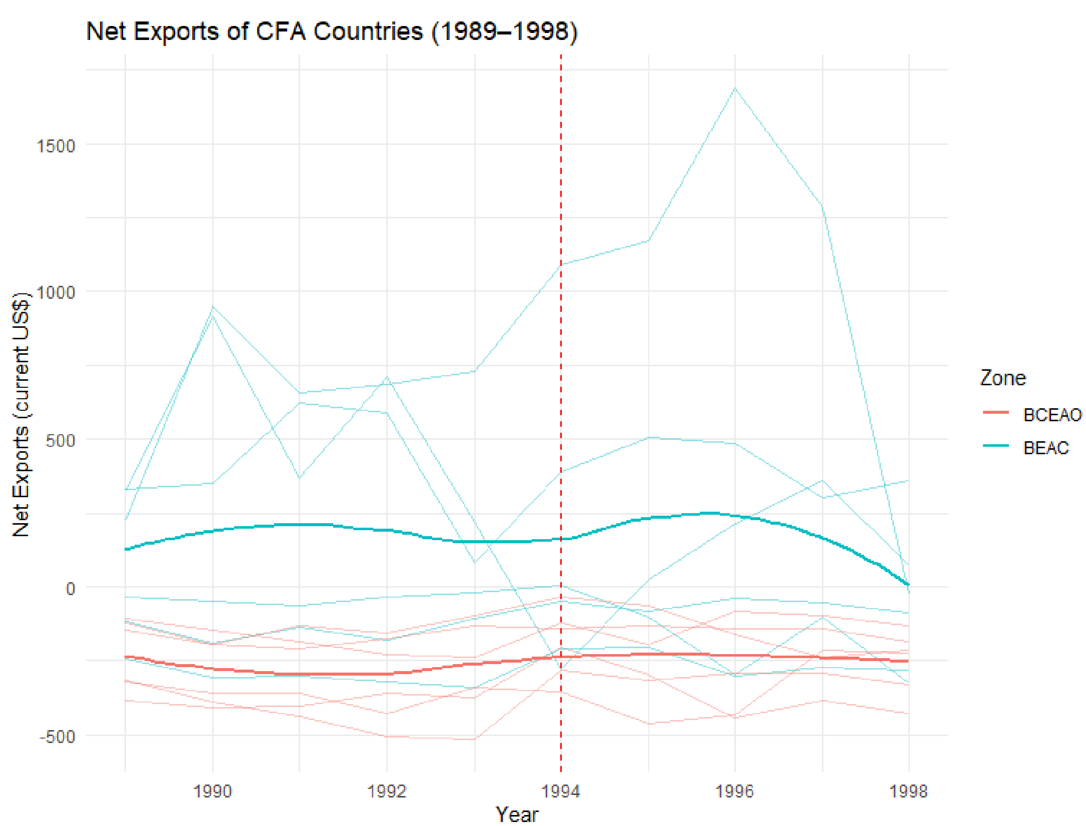
IV. Results

A. Descriptive Statistics and Visualizations

(1) Net Exports

(a) In the BCEAO zone, average net exports improved (deficit narrowed) from –278.5 billion US$ pre-devaluation to –235.2 billion post-devaluation, which is a gain of about 43.3 billion. BEAC experienced a small drop from 176.8 billion to 174.5 billion, suggesting negligible trade-volume response overall

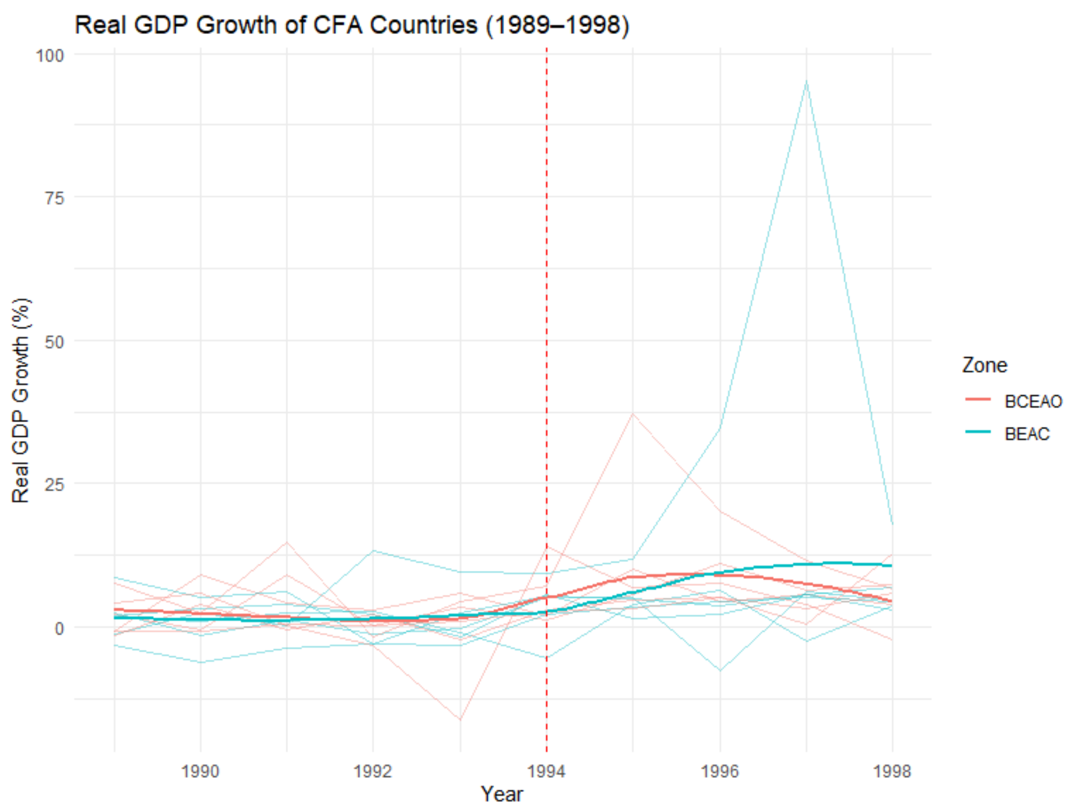
(b) Both unions show highly variable country‑level series around 1994. BCEAO's aggregate trend line gradually moves upward post-devaluation, reflecting a narrowing deficit, although individual countries differ in timing and magnitude. BEAC exhibits a slight downward tilt after 1995, indicating that central African exporters saw only minor improvements in trade balance.



(2) Real GDP Growth:

(a) Both zones saw substantial post-devaluation growth. BCEAO's average growth jumped from 1.85% to 7.21% (an increase of 5.36 percentage points), while BEAC rose from 1.36% to 8.41% (an increase of 7.05 percentage points), reflecting a strong expansionary impulse.

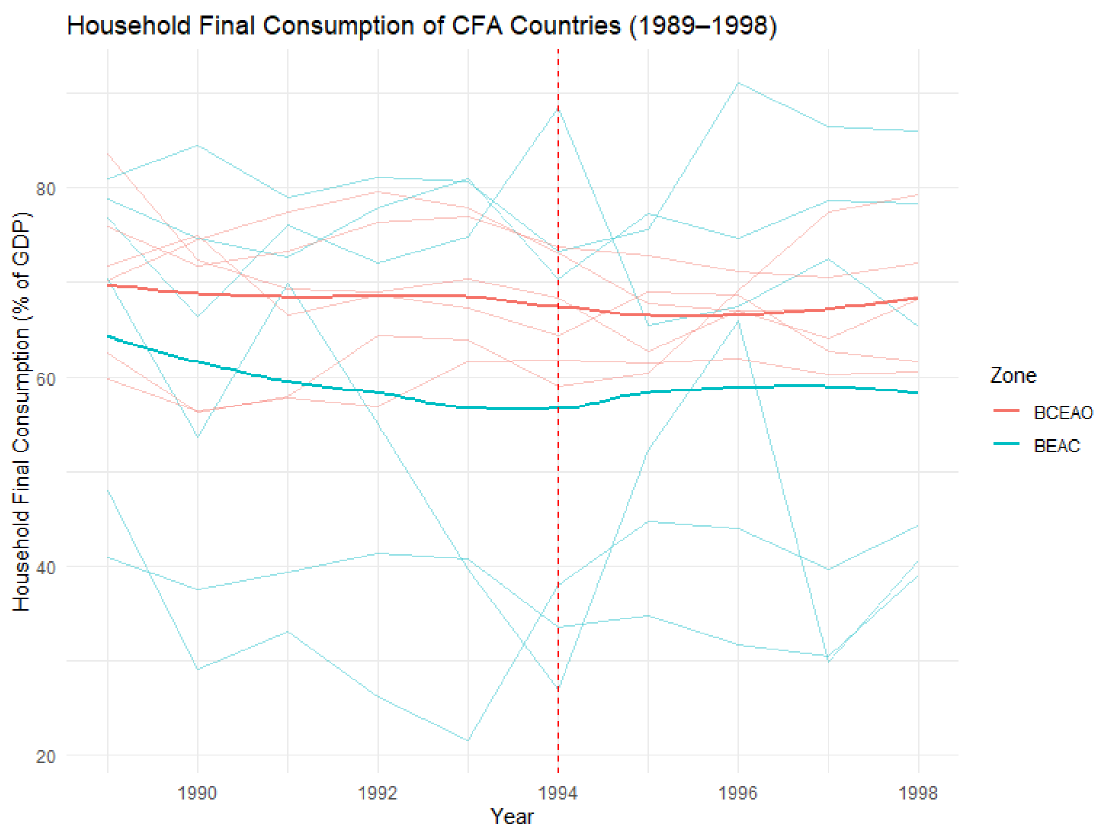
(b) The dashed 1994 line marks a clear turning point: before the devaluation, growth rates hover around 1–2% annually, but post-1994 both BCEAO and BEAC trend sharply higher, reaching peaks near 8–10%. This jump is more pronounced in BEAC, suggesting stronger output responsiveness in central Africa.



(3) <u>Household Consumption:</u>

(a) Private consumption fell modestly after the devaluation: BCEAO's ratio declined by 1.77 percentage points (from 68.83% to 67.06%), and BEAC's dropped by 1.90 percentage points (from 60.11% to 58.21%). This suggests limited or delayed pass-through to household spending.

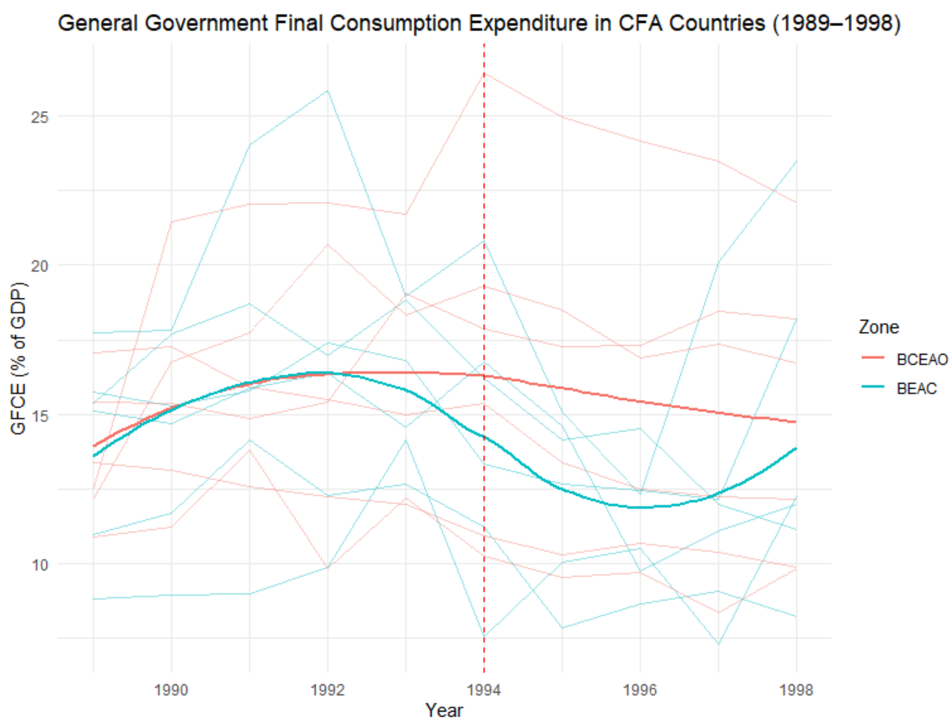
(b) Private consumption shares are fairly flat leading up to 1994, then drift slightly downward in both zones. The smoothed lines show a modest decline of 1–2 percentage points after devaluation, indicating that households did not immediately increase spending despite improved output.



(4) <u>Government Consumption:</u>

(a) Government consumption in BCEAO edged down by 0.11 percentage points (15.59% to 15.48%), whereas BEAC saw a larger decrease of 2.55 percentage points (15.40% to 12.85%), pointing to fiscal tightening or reallocation in the post-devaluation period.

(b) Government consumption for BCEAO dips marginally around 1994 before recovering to a stable post-devaluation level, whereas BEAC's public spending line declines more steadily after the devaluation. The contrasting slopes imply that West African governments may have paused or reallocated spending briefly, while central African budgets tightened more persistently.



B. Variance Analysis

Zone	Var_NetX	Var_GDP	Var_HC	Var_GC
BCEAO	15576.59	46.45	43.38	19.65
BEAC	211360.22	178.58	416.34	17.32

The pronounced dispersion in the BEAC series suggests notable volatility; we now quantify this variance by zone.

(1) Net Exports: BEAC's variance in net exports considerably exceeds BCEAO's, reflecting more volatile trade balances.

(2) GDP Growth: Variance in annual growth for BEAC is higher than for BCEAO, indicating output responses are more heterogeneous in central African countries.

(3) Household Consumption: Household consumption exhibits a larger variance in BEAC, confirming the greater dispersion seen in the smoothed trend lines.

(4) Government Consumption: Government consumption exhibits a marginally higher variance in BCEAO compared to BEAC.

Higher volatility in BEAC countries can likely be attributed to the differences in the economies of BEAC and BCEAO countries. BEAC countries are resource-rich and tend to rely more heavily on oil and mineral exports, making trade revenues and growth highly sensitive to global price swings. Such fluctuations in prices generate large swings in revenues and growth across the BEAC member states, leading to higher variance in macro indicators. On the other hand, BCEAO countries are predominantly agricultural and, therefore, face milder commodity‑price volatility, resulting in tighter clustering of outcomes across countries

C. Welsh t-test

Variable	Mean_Pre	Mean_Post	MeanDiff	Statistic	PValue
NetExports	-30.34	-50.83	-20.49	0.28	0.78
Real GDP growth (annual %)	7.81	1.61	-6.20	3.35	0.001
Household final consumption expenditure (% of GDP)	62.64	64.47	1.84	-0.64	0.53

General government final consumption expenditure (% of GDP)	14.16	15.49	1.33	-1.69	0.09

(1) Net Exports: Post-devaluation net exports declined by about 20.5 billion USD, but this change is not statistically significant ($t = 0.28$, $p = 0.78$), suggesting any reduction may stem from noise or external shocks.

(2) Real GDP Growth: Average GDP growth fell by approximately 6.2 percentage points, a significant decline ($t = 3.35$, $p = 0.001$), contrary to the predicted expansionary effect of devaluation.

(3) Household Consumption: Household consumption increased by around 1.8 percentage points; however, this rise is not statistically significant ($t = -0.64$, $p = 0.53$), indicating minimal impact on private spending.

(4) Government Consumption: Government consumption rose by roughly 1.3 percentage points. This increase is marginally significant at the 10% level ($t = -1.69$, $p = 0.093$), pointing to a modest post-devaluation fiscal adjustment.

V. Conclusion

Our results offer a nuanced view of the Mundell–Fleming predictions. While BEAC displays the expected improvement in net exports, BCEAO’s muted response indicates region-specific factors—such as import dependency, global commodity price movements, or infrastructural bottlenecks—that may dampen pass-through. The robust GDP growth response underscores the potency of exchange-rate adjustments in fixed regimes, although the magnitude varies by union, suggesting differences in economic structure and openness. The stability of household consumption despite a stronger trade balance implies that real income gains did not uniformly translate into private spending, possibly due to financial frictions or uneven distributional effects. Divergent government consumption patterns point to heterogeneity in fiscal frameworks: BCEAO governments may have leveraged devaluation to support domestic spending, while BEAC counterparts pursued fiscal restraint, potentially under external conditionality. Our limitations include the use of annual data, so effects happening within a year may be hidden, and the comparison of just two currency unions without a broader control group. We also omit global shocks like oil price swings. Future research should use higher-frequency data and include additional comparison groups.

Empirical results challenge the textbook Mundell–Fleming predictions in several ways. Real GDP growth fell by 6.2 percentage points post-devaluation ($t = 3.35$, $p = 0.001$), instead of rising, suggesting that higher import costs or external shocks may have overwhelmed the trade stimulus. Net exports declined by 20.5 billion USD (not significant, $p = 0.78$) and household consumption rose only 1.8 percentage points (not significant, $p = 0.53$), indicating limited response in trade volumes and private demand. Government consumption increased by 1.3 percentage points (marginally significant, $p = 0.093$), pointing to a modest expansion of public spending rather than consolidation.

These findings highlight the importance of external shocks and elasticities, as global commodity price movements and inelastic trade volumes can mute the impact of currency devaluations. In addition, institutional and structural heterogeneity is important because differences in fiscal capacity, import dependence, and policy frameworks across BCEAO and BEAC shape divergent outcomes.

For policymakers, this points to the importance of pairing devaluation with financial market reforms and export diversification to strengthen the trade channel. They should also use targeted fiscal measures—rather than blanket spending cuts or increases—to support vulnerable sectors. In the future, it would be valuable to estimate country-level models with controls for global price shocks, employ quarterly or monthly data to capture short-run dynamics, and introduce a broader control group of non-CFA countries for difference-in-differences analysis.

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Speech by Kwame Nkrumah at the summit of the Organisation of African Unity.